

Applications of Electrolysis

Contents

- * Electroplating
- * Hydrogen Fuel Cells

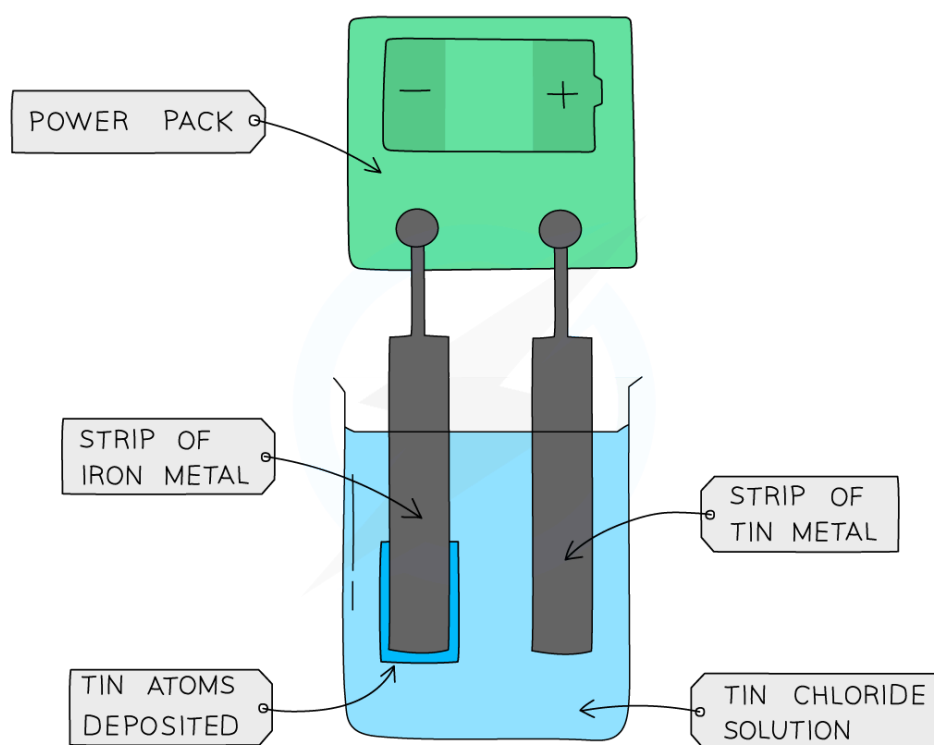


Electroplating

What is electroplating?

- Electroplating is a process where the surface of one metal is **coated** with a layer of a different metal
- The **cathode / negative electrode** is the **object** to be electroplated
- The **anode / positive electrode** is made from the **pure** metal that will be plated onto the object
- The **electrolyte** is an **aqueous solution** of a soluble salt of the pure metal at the anode

Electroplating a strip of iron with tin



Copyright © Save My Exams. All Rights Reserved



A piece of iron being electroplated with tin. The electrolyte is tin(II) chloride, a water-soluble salt of tin

- At the anode:
 - Tin atoms lose electrons to form tin ions in solution



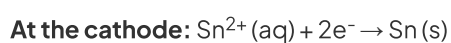
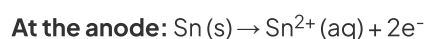
Your notes

- The loss of electrons is oxidation
- At the cathode:
 - Tin ions gain electrons to form tin atoms
 - The gain of electrons is reduction
 - The tin atoms are deposited on the strip of iron metal, coating it with a layer of tin



Examiner Tips and Tricks

Extended Tier students may be asked to write the ionic half equations for the reaction at each electrode. For the example above, these would be:



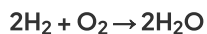
What is electroplating used for?

- Electroplating is done to make metals more **resistant** to corrosion or damage
 - e.g. chromium and nickel plating
 - e.g. galvanising - zinc plating / coating, typically done to iron and steel
- It is also done to **improve the appearance** of metals,
 - e.g. coating cutlery and jewellery with silver



Hydrogen fuel cells

- A fuel is a substance which releases energy when burned
- Hydrogen is used as a fuel in rocket engines and in fuel cells to power some cars
- A fuel cell is an electrochemical cell in which a fuel **donates** electrons at one electrode and oxygen **gains** electrons at the other electrode
 - $\text{H}_2 \rightarrow 2\text{H}^+ + 2\text{e}^-$
 - $\text{O}_2 + 4\text{H}^+ + 4\text{e}^- \rightarrow 2\text{H}_2\text{O}$
- The **hydrogen-oxygen** fuel cell produces electricity by combining both elements, releasing energy and water
- The overall equation for the reaction within a hydrogen fuel cell is:



- The diagram below shows the setup of a hydrogen fuel cell
 - The air entering provides the oxygen
 - The fuel entering is hydrogen
 - The only chemical product made is **water**



Your notes

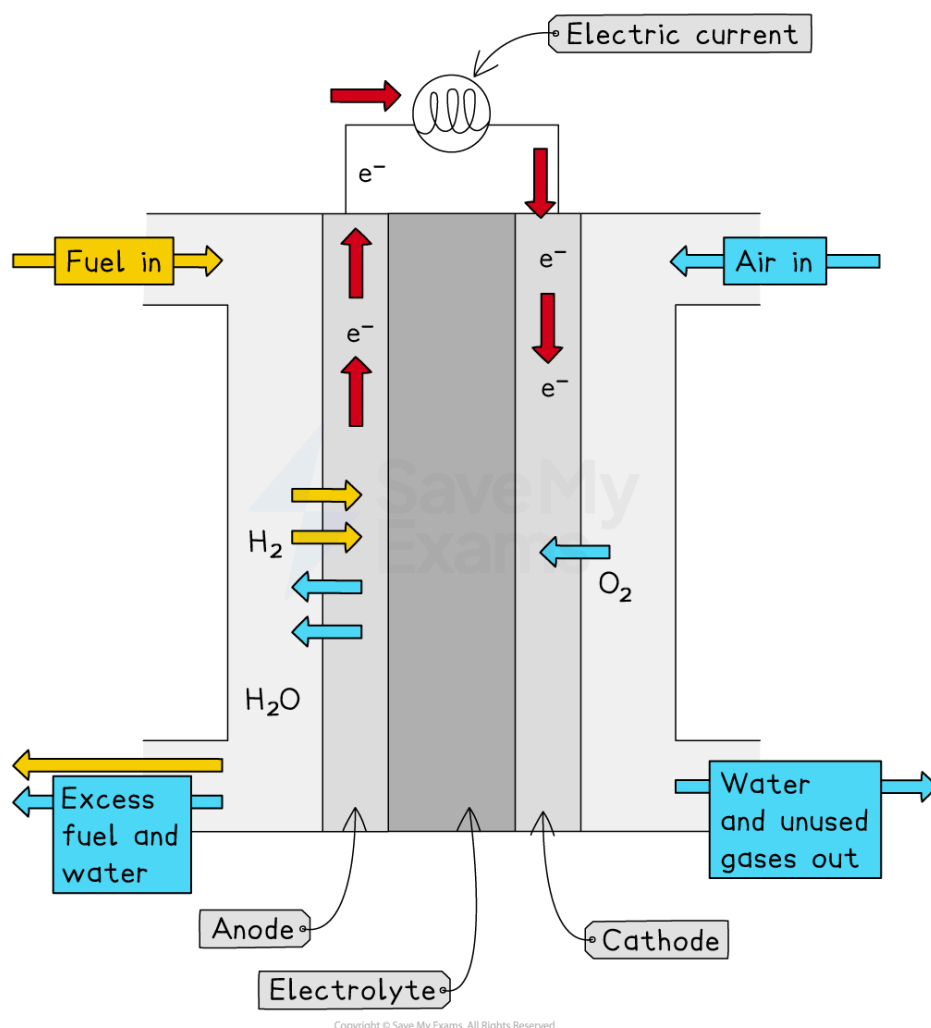


Diagram showing the movement of hydrogen, oxygen and electrons in a hydrogen-oxygen fuel cell

Advantages & disadvantages of hydrogen fuel cells

Extended tier only

- Hydrogen-oxygen fuel cells are an alternative technology for powering vehicles, replacing traditional petrol or diesel engines.

Advantages

No pollutants produced

- The only product of the reaction is water. This is a major advantage over petrol engines which produce harmful pollutants such as carbon dioxide (a greenhouse gas), oxides of nitrogen (which cause acid rain), and carbon monoxide.

High efficiency

- They release more energy per kilogram than either petrol or diesel.

Renewable

- The hydrogen fuel can be produced by the electrolysis of water, which is a renewable resource.

Disadvantages

Hydrogen production:

- Most hydrogen is currently produced from fossil fuels (like methane), which is a non-renewable process that releases carbon dioxide.
- The electrolysis of water requires large amounts of electricity, which may be generated by burning fossil fuels.

Storage and Safety

- Hydrogen is a highly flammable gas that is difficult and expensive to store safely, often requiring high-pressure tanks. This is less convenient than storing liquid petrol.

Cost and Infrastructure:

- The materials used to make fuel cells are often expensive.
- There is a very limited number of hydrogen filling stations available.



Examiner Tips and Tricks

When comparing fuel cells to petrol engines, your answers should focus on the **chemical products** and their **environmental impact**.

Fuel cells are much quieter, leading to less noise pollution. However, this is considered a physical or engineering advantage. So, this is unlikely to score a mark.

Always stick to the chemical points, like "**the only product is water**" or "**no carbon dioxide is produced**."



Your notes